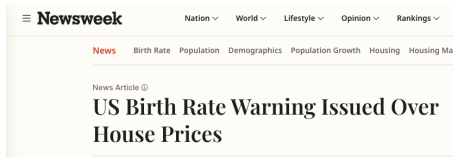


Fertility and Housing Tenure Choice

Tommaso De Santo

September 14, 2026

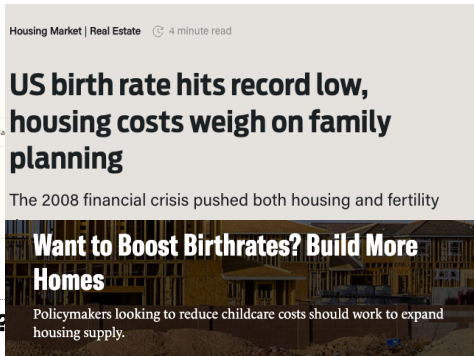
Housing and Fertility



POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.



Renting a home 'leads to lower birth rates'

Housing, Fertility, and Tenure Choice

Housing costs affect fertility, but tenure and wealth matter:

- Earlier access to homeownership increases family size (Fazio et al. 2025)
- Credit access raises buying, births, and home size (Hacamo 2021, Couillard 2025)

Children increase how big of a house you need. Key channels:

- Owning: tenure segmentation (big houses are often only for sale) [▶ Data](#)
- Wealth: borrowing constraints affect purchase ability
- Intergenerational distribution: old people own most of the large dwellings [▶ Data](#)

How much can physical space affect fertility?

What kind of housing policy is good for natality?

This Paper

Question: how much does access to space shape fertility, and can housing policy change it?

Model: a lifecycle economy with one housing market, in which children need space

- Households choose when to try for children; children raise the space they need.
- Renting is capped in size: large homes must be bought (tenure segmentation).
- Buying requires a down payment: wealth and its timing matter (borrowing constraints).
- Households age, save, and bequeath: large homes end up with the old (intergenerational allocation).

Quantification: the U.S. as a transition to a smaller population

- Calibrate the 2007 economy; fit the 2007–2023 fertility decline with preference shocks.
- Treat 2023 as a point on the transition and project forward.
- Policy: can a property tax that shifts space to young families raise births?

Model

Environment

- Discrete time, stationary overlapping generations of households
- Adult age $a \in [0, J]$ with survival probability s_a ; working age $a \in [0, a_R]$; fertile window $a \in [a_F^0, a_F^1]$
- Children:
 - mature stochastically; housing need depends on children at home
 - are dependents and do not work
- Households differ in financial wealth, inherited housing product, family composition, and persistent idiosyncratic labor efficiency z_t
- Households choose:
 - **Fertility**: whether to attempt one additional birth, subject to EV shocks
 - **Tenure**: rent or choose a product from a discrete owner menu, subject to EV shocks
 - **Savings**: next-period liquid wealth b_{t+1} ; mortgages secured by the house

Preferences

n_t : children ever born; m_t : children currently at home.

Flow utility. The consumption–space composite is

$$C_t = \frac{c_t^{\alpha_0} (s_t^{\text{net}})^{1-\alpha_0}}{e(m_t)}, \quad u_t = \frac{C_t^{1-\sigma}}{1-\sigma} + \psi_t m_t^{1-\gamma}, \quad \gamma \in [0, 1).$$

- α_0 : consumption share; σ : utility curvature; γ : curvature of the child benefit.
- $e(m_t) = ((2 + \omega m_t)/2)^\nu$; ω weights children and ν governs household economies of scale.
- $\psi_t m_t^{1-\gamma}$ is the direct benefit; a successful first birth carries the one-time utility cost ξ .

Housing services. With children at home, the floor is h_P ; χ scales owners' residual space:

$$s_t^{\text{net}} = \mathbf{1}\{\text{own}\} \chi [h_t - h_P \mathbf{1}\{m_t > 0\}] + \mathbf{1}\{\text{rent}\} [h_t^R - h_P \mathbf{1}\{m_t > 0\}] > 0, \quad \chi > 0.$$

Bequests. With $w_{t+1}^e = b_{t+1} + P_t h_t$, terminal utility is normalized to zero at a zero estate:

$$B(w_{t+1}^e) = \theta_0 \frac{(\theta_1 + \max\{w_{t+1}^e, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1-\sigma}.$$

Income and saving.

- After-tax earnings $y_t^d(a, z_t)$ and a lump-sum transfer T_t .
- One-period bond b_{t+1} at gross rate R_b ; owners borrow only against the house.

Owning. Choose h_t from a discrete owner-product menu, with largest size $\bar{h}$, arriving with inherited h_{t-1} .

- The house depreciates at rate δ_H and is taxed at rate τ_t^P .
- Selling costs a fraction ψ^s of the sale value; ϕ is the financed share.
- Let $S_t = (1 - \psi^s)P_t h_{t-1}$ in a transaction and $S_t = 0$ otherwise. A change to an owner product requires

$$R_b b_t + S_t + y_t^d + T_t \geq (1 - \phi)P_t h_t, \quad b_{t+1} \geq -\phi P_t h_t.$$

Renting. Rent $h_t^R \in [0, \bar{h}^R]$ at rent r_t per unit, with $\bar{h}^R < \bar{h}$.

- The largest homes are reachable only through ownership.

Saving and Sale Solvency

Renters have no unsecured credit, and selling to rent must settle inherited debt:

$$b_{t+1} \geq 0, \quad R_b b_t + S_t \geq 0 \quad \text{when an owner sells to rent.}$$

Current income can finance a purchase; it cannot cover a shortfall when selling to rent.

An owner retaining the same home can roll over debt beyond the collateral limit, but cannot increase that excess debt:

$$b_{t+1} \geq \min\{b_t, -\phi P_t h_t\}.$$

When death is possible, the estate after selling costs must be nonnegative:

$$b_{t+1} \geq \max\{\min(b_t, -\phi P_t h_t), -(1 - \psi^s) P_t h_t\}.$$

Fertility and Child Aging

- During fertile ages, households choose each period whether to attempt at most one additional birth; children ever born are capped at $3+$.
- The choice is discrete, with an extreme-value taste shock ε_t^F of scale κ_1 for the first birth and κ_C for later births.
- An attempt succeeds with age-specific probability π_a and adds one child.
- Children mature stochastically: each dependent child leaves the household with a fixed probability each period, independently of the others.

Firms and Earnings

- Competitive firms produce with effective labor:

$$Y_t = AL_t, \quad w = A, \quad L_t = \int_{a < a_R} e_a z_t dG_t.$$

- Age- and productivity-dependent earnings:

$$y_t^d(a, z_t) = \begin{cases} (1 - \tau^w) w e_a z_t, & a < a_R, \\ \varpi_t, & a \geq a_R. \end{cases}$$

- e_a : age-earnings profile; z_t : idiosyncratic labor efficiency; G_t : distribution of households over states.
- Retirees receive a pay-as-you-go pension ϖ_t ; payroll taxes balance it each period.

Housing Supply and Rents

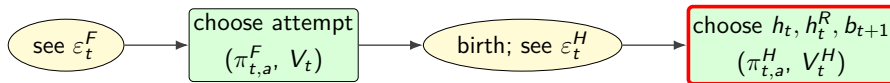
The stationary supply schedule uses the current user cost:

$$H^s = H_0 \left(\frac{r}{\bar{r}} \right)^\eta, \quad r = (R_b - 1 + \delta_H + \tau_p)P.$$

- H_0 : housing supply at reference rent $\bar{r}$; η : supply elasticity.
- More generally, a landlord's user-cost identity is

$$r_t = (R_b + \delta_H + \tau_t^p)P_t - P_{t+1}.$$

The Household's Problem: Housing and Savings



Conditional on the realized family size, choose housing and savings:

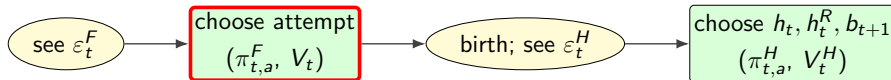
$$V_t^H(b_t, h_{t-1}, z_t, a, n, m) = \mathbb{E}_{\varepsilon_t^H} \max_{h_t, h_t^R, b_{t+1}} \left\{ u_t(c_t, s_t^{\text{net}}; m) + \varepsilon_t^H(h_t) \right. \\ \left. + \beta \mathbb{E}_t[s_a V_{t+1}(b_{t+1}, h_t, z_{t+1}, a+1, n_{t+1}, m_{t+1}) + (1-s_a)B(w_{t+1}^e)] \right\}.$$

subject to

$$c_t + b_{t+1} + r_t h_t^R + (\delta_H + \tau_t^p) P_t h_t = R_b b_t + y_t^d(a, z_t) + T_t + S_t - \mathbf{1}\{h_t \neq h_{t-1}\} P_t h_t,$$

with the financing, solvency, and housing-service constraints above.

The Household's Problem: Fertility Choice



Waiting and attempting have values

$$W_t^0 = V_t^H(b_t, h_{t-1}, z_t, a, n_t, m_t),$$

$$W_t^A = \pi_a [V_t^H(b_t, h_{t-1}, z_t, a, n_t + 1, m_t + 1) - \xi \mathbf{1}\{n_t = 0\}] \\ + (1 - \pi_a) V_t^H(b_t, h_{t-1}, z_t, a, n_t, m_t).$$

Before fertility tastes, household value is

$$V_t(b_t, h_{t-1}, z_t, a, n_t, m_t) = \mathbb{E}_{\varepsilon_t^F} \max\{W_t^0 + \varepsilon_{t,0}^F, W_t^A + \varepsilon_{t,A}^F\}.$$

Choice probabilities:

$$\pi_{t,a}^F(b_t, h_{t-1}, z_t, n_t, m_t) = \frac{\exp\{W_t^A/\kappa(n_t)\}}{\exp\{W_t^A/\kappa(n_t)\} + \exp\{W_t^0/\kappa(n_t)\}},$$

Here $\kappa(n_t) = \kappa_1$ if $n_t = 0$ and κ_C otherwise. Expected births equal $\pi_a \pi_{t,a}^F$.

Stationary Equilibrium

At stationary prices, optimal household choices and survival reproduce the distribution of adult households:

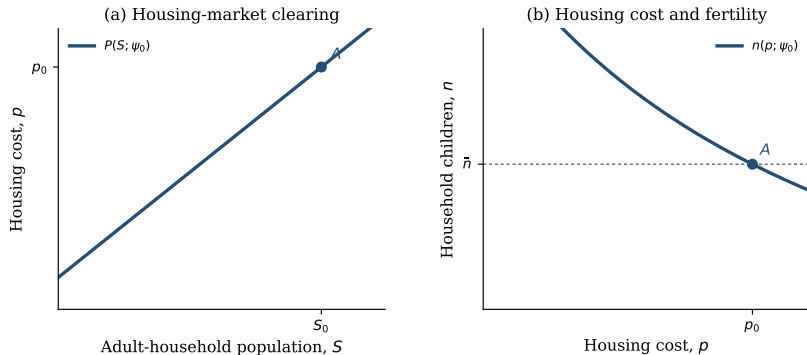
$$G' = G, \quad E = \frac{B^{\text{adj}}}{2.1}, \quad B^{\text{adj}} = B + (w_{3+} - 3)B_{3+}.$$

- The adjusted-birth flow feeds an entry queue: half enter at age 16 and half at age 20; adult survival and these entrants reproduce G .
- E : adult-household entry; B : births; B_{3+} : births entering the 3+ state; w_{3+} : its representative number of children.
- Housing clears, $H^s = H^d$, and the PAYGO pension budget balances. Property-tax receipts are not rebated to households ($T = 0$).

Quantification

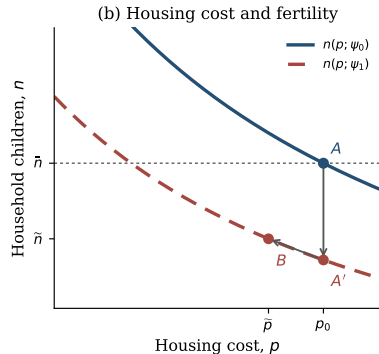
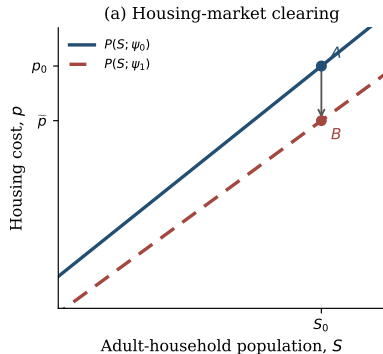
Initial Steady State

For intuition, summarize housing costs by p and fertility by $n(p; \psi)$.



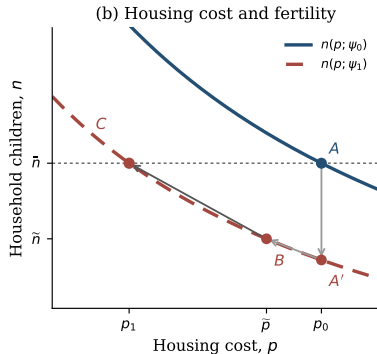
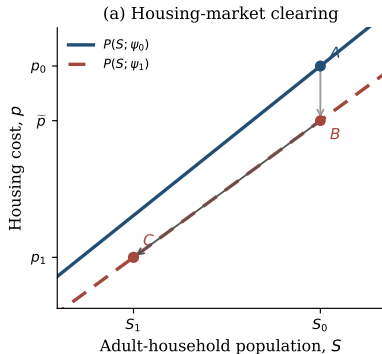
At A , housing clears at (S_0, p_0) and fertility is at the replacement rate $\bar{n}$.

Impact



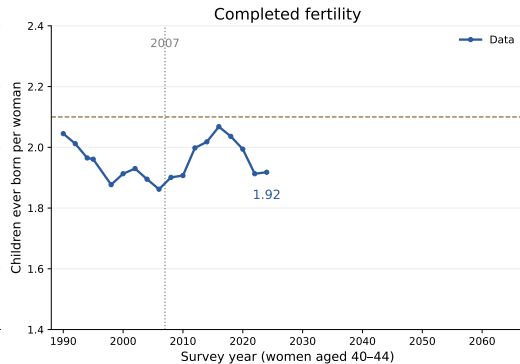
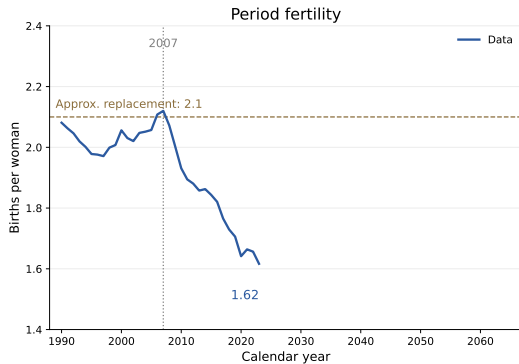
A fall from ψ_0 to ψ_1 lowers fertility at unchanged housing cost: $A \rightarrow A'$. Housing costs then adjust at inherited population S_0 : $A' \rightarrow B$.

Demographic Adjustment

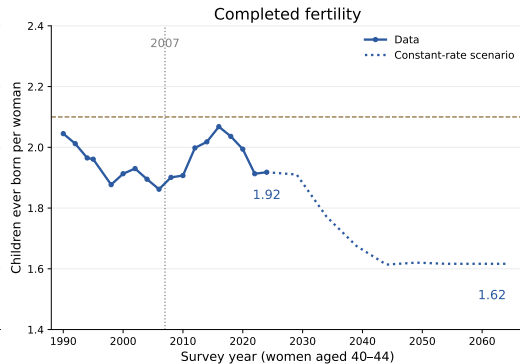
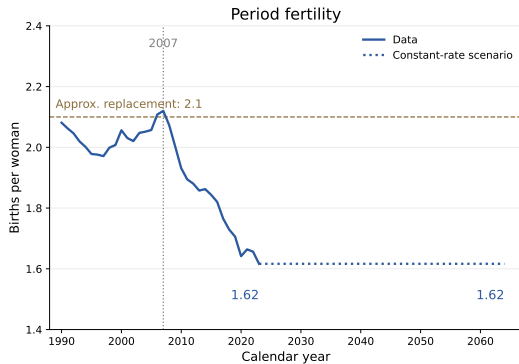


Smaller entering cohorts reduce housing demand. Lower housing costs partly offset the preference decline; C marks the new stationary outcome.

Fertility in the United States



Fertility in the United States



1. Calibrate the 2007 steady state.

Match fertility, housing, and wealth moments to determine household parameters.

2. Estimate the sequence of fertility shocks.

Hold those parameters fixed and match birth rates through 2023 with unanticipated changes in fertility preferences. Carry households and population forward between shocks.

3. Project the economy forward.

After 2023, hold fertility preferences fixed and solve for prices, housing choices, and population. Compare housing policies from the same 2023 economy.

A model period is four calendar years.

Utility Specification

Utility depends on consumption and space above the parenthood requirement:

$$u_t(c_t, s_t^{\text{net}}; m_t) = \frac{\left[\frac{c_t^{\alpha_0} (s_t^{\text{net}})^{1-\alpha_0}}{e(m_t)} \right]^{1-\sigma}}{1-\sigma} + \psi_t m_t^{1-\gamma}, \quad \gamma \in [0, 1).$$

$$e(m_t) = \left(\frac{2 + \omega m_t}{2} \right)^\nu, \quad \bar{h}(m_t) = h_P \mathbf{1}\{m_t > 0\}.$$

$$s_t^{\text{net}} = \mathbf{1}\{\text{own}\} \chi [h_t - \bar{h}(m_t)] + \mathbf{1}\{\text{rent}\} [h_t^R - \bar{h}(m_t)] > 0.$$

The physical floor is removed before the owner-service multiplier χ ; $e(m_t)$ follows Scholz, Seshadri & Khitatrakun (2006).

Identification

Ten parameters are chosen jointly to match twelve moments.

Fertility

- ψ_0 , preference for children $\rightarrow$ completed fertility
- ξ , first-birth cost $\rightarrow$ childlessness
- κ_1 , first-birth taste dispersion $\rightarrow$ mean age at first birth, first births after 30
- κ_C , later-birth taste dispersion $\rightarrow$ one-child families

Housing

- h_P , space requirement of parents $\rightarrow$ rooms added at first birth, rooms of 3+ versus 1–2 children
- χ , owner premium $\rightarrow$ ownership rate
- H_0 , supply scale $\rightarrow$ mean rooms

Saving and bequests

- β , discount factor $\rightarrow$ wealth over earnings
- θ_0 , bequest scale $\rightarrow$ bequest flow
- θ_1 , bequest wealth shift $\rightarrow$ old-age p90/median

Calibration: Targets and Model

Moment	Target	Model
Completed fertility	2.10	2.10
Childless, women 40–44 (%)	19.8	20.7
Exactly one child among mothers, 40–44 (%)	21.4	20.7
Mean age at first birth (years)	25.98	26.10
First births at age 30+ (%)	24.9	23.2
Mean occupied rooms (capped at 9)	5.56	6.29
Ownership, heads 30–55 (%)	64.8	52.4
First-birth room response, –1 to +3	0.720	1.011
Rooms: 3+ versus 1–2 resident children	0.347	0.200
Wealth / annual gross earnings	6.15	4.85
Annual bequests / wealth (%)	0.88	0.86
Wealth/income p90 / median, ages 76–84	3.52	4.44

Sources: CPS 2004/2006 (children ever born, women 40–44); natality records 2003–2006 (first-birth timing); ACS 2005/2006, 42 metros (rooms, ownership); PSID 2003/2005 (wealth); PSID event study (room response).

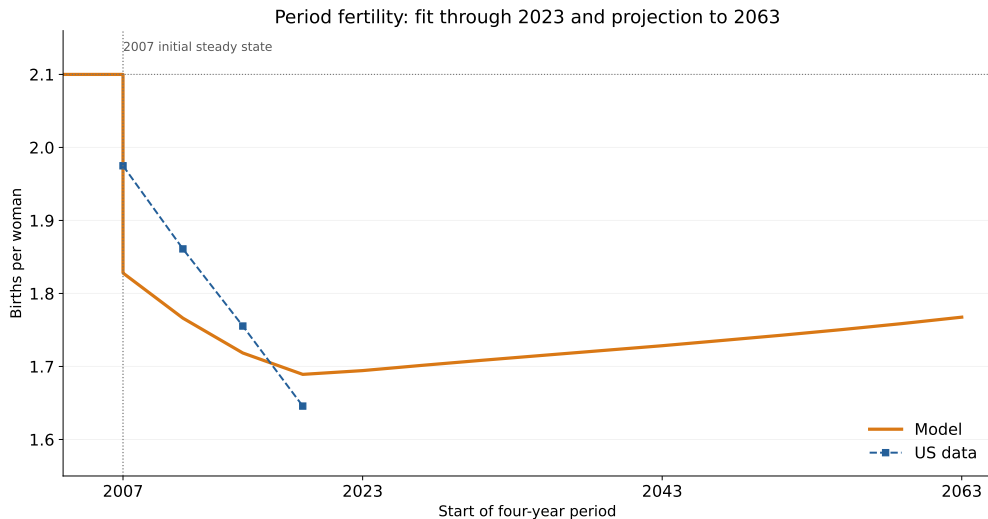
Internally Calibrated Parameters

Parameter	Economic interpretation	Value
β_{annual}	Annual discount factor	0.990
κ_1	First-birth taste dispersion	0.279
κ_C	Later-birth taste dispersion	0.347
χ	Owner housing-service multiplier	1.048
H_0	Housing supply scale	8.191
θ_0	Bequest-motive scale	0.087
θ_1	Bequest-utility wealth shift	0.057
ξ	First-birth utility cost	0.127
h_P	Parenthood space requirement	2.300
ψ_0	Initial preference for children	0.160

Externally Calibrated Parameters

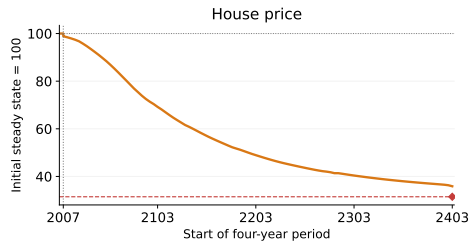
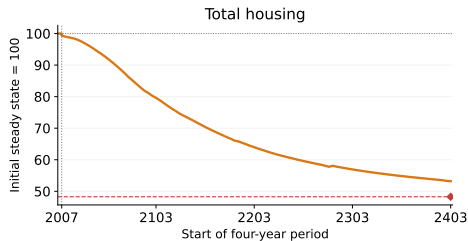
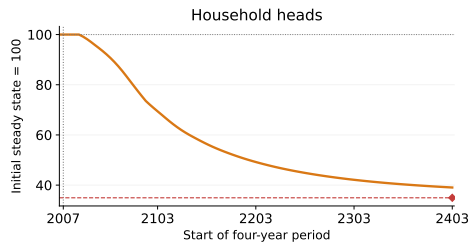
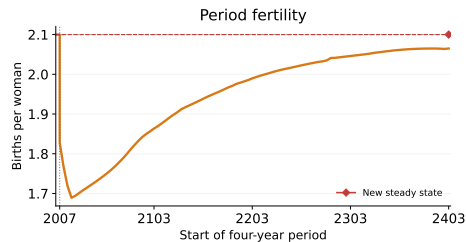
Parameter	Value	Source
Risk aversion σ	2	
Equivalence scale (ω, ν)	(0.7, 0.7)	Scholz et al. (2006)
Consumption share α_0	0.733	CEX 2019–23, childless renters
Housing-supply elasticity η	0.63	Baum-Snow & Han (2024)
Annual real interest rate q_{ann}	2.0%	Greaney et al. (2025)
Payroll tax rate τ^w	17.9%	OECD U.S. average
Financed share ϕ	0.80	20% down payment
Depreciation δ_H , property tax τ^p	2%, 1% per year	
Selling cost ψ^s	6%	Greaney et al. (2025)
Rental size cap $\bar{h}^R$	6 rooms	Greaney et al. (2025)
Ages	18, 66, 45	entry, retirement, last fertile age

Fertility Transition: 2007–2063



Full Transition: 2007–2403

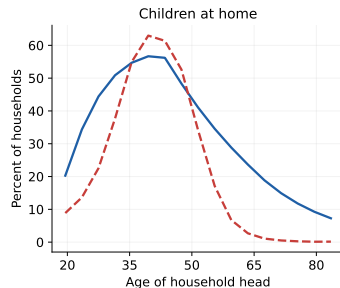
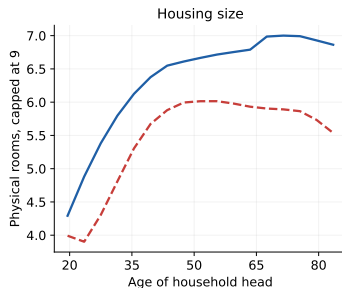
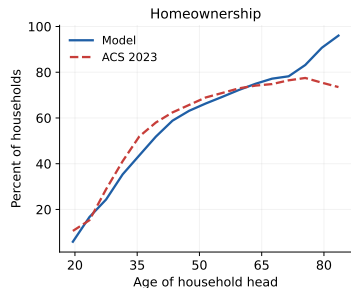
Full transition after a permanent fertility-preference decline



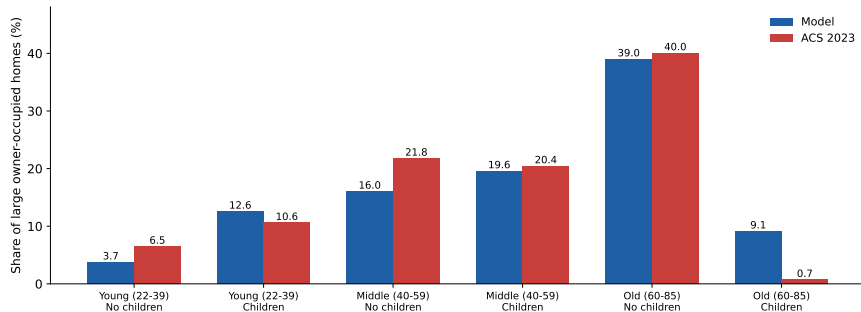
2023 Data and Model

Moment	Data	Model
Completed fertility, ages 40–44	1.92	1.70
Childless, ages 40–44 (%)	18.8	25.0
Exactly one child among mothers, 40–44 (%)	22.2	25.5
Mean age at first birth (years)	28.09	29.74
First births at age 30+ (%)	37.7	44.0
Mean occupied rooms (capped at 9)	5.58	6.07
Ownership, heads 30–55 (%)	58.7	46.9
First-birth room response, –1 to +3	0.720	1.060
Rooms: 3+ versus 1–2 resident children	0.355	0.266
Wealth / annual gross earnings	6.87	5.64
Annual bequests / wealth (%)	0.88	0.70
Wealth/income p90 / median, ages 76–84	3.45	3.50

Lifecycle Fit in 2023



Intergenerational Allocation: Model vs Data

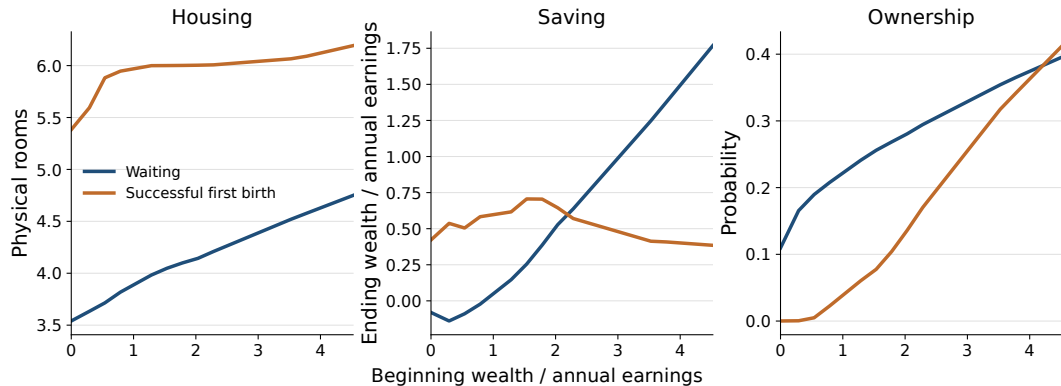


Share of owner-occupied homes with 6+ rooms; ACS 2023.

- Large homes sit with older households without children; young families hold few.
- Coven et al. (2025): a rebated property tax moves space from old owners to young households.
- Those young households are the ones deciding on children, and children need space.

Household Choices at First Birth

Initially childless renters, ages 22–25, at the middle earnings state.

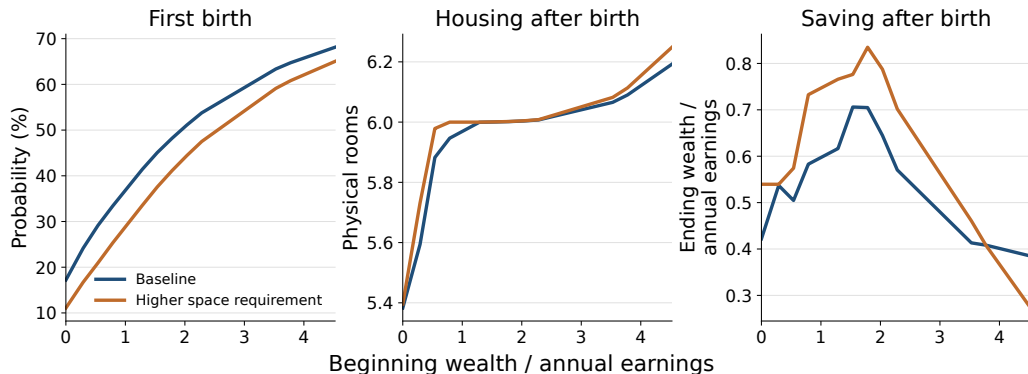


A first child raises space demand; at low wealth, households rent and retain more liquid assets.

The Parenthood Space Requirement

Initially childless renters, ages 22–25, middle earnings state.

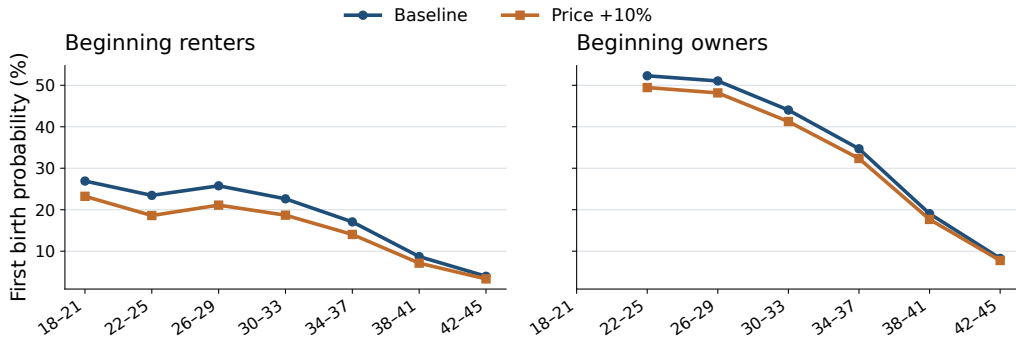
Raise required space from 2.594 to 2.781 rooms; prices and other inputs stay fixed.



Requiring more space reduces first-birth probabilities at the same household resources.

Housing Costs and Fertility

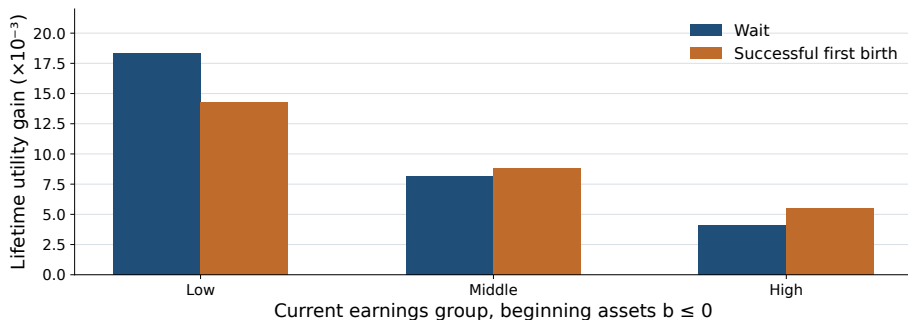
Impose a 10% increase in house prices and rents; evaluate households at the same initial states.



Birth probabilities fall, with a larger response among households that start as renters.

Mortgage Finance and Fertility

Increase the financed share from 80% to 95%, holding house prices and rents fixed.



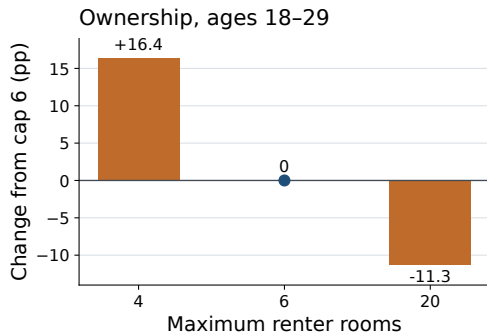
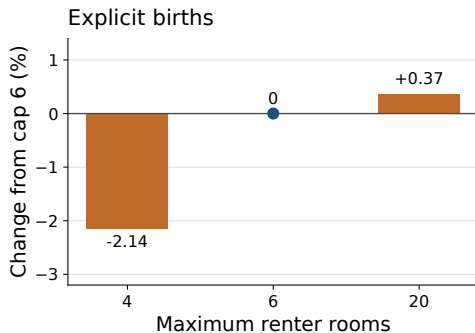
Value gains where both choices have positive probability: 92.5% of low-earner exposure.

Credit helps both choices; fertility depends on the gain from a child relative to waiting.

Across the lifecycle: young ownership +10.1 percentage points; birth flows -0.25% .

Rental Space and Homeownership

Change the maximum size of rental housing, holding prices and other inputs fixed.



Larger rentals substitute for ownership, while birth flows change little.

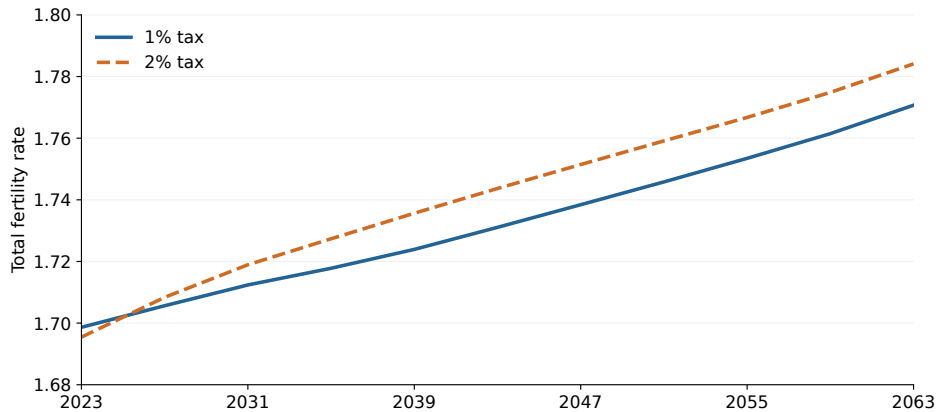
The experiment of Coven, Golder, Gupta & Ndiaye (2025)

- Tax housing by value; rebate the revenue equally to all households.
- Older owners pay more than they get back; young renters get back more than they pay.
- House prices fall; constrained young households buy sooner or buy bigger.

This paper

- Same experiment: property tax from 1% to 2%, equal rebate, from the 2023 economy.
- Follow prices, rents, who lives in the large homes, and births along the path.
- Does the space that moves to young families turn into children?

Policy Results



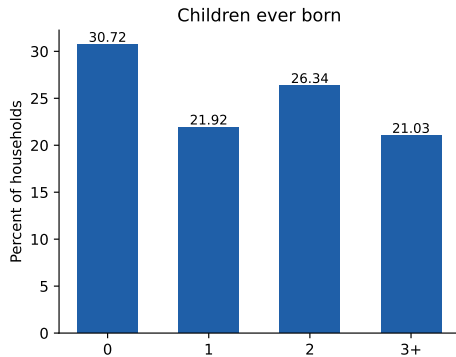
Conclusion

- A lifecycle model of fertility with realistic housing: tenure, down payments, space
- Children need space; large homes must be bought; the old hold most of them
- Today is a point on a transition to a smaller population
- A rebated property tax can move space toward young families, generating a fertility response

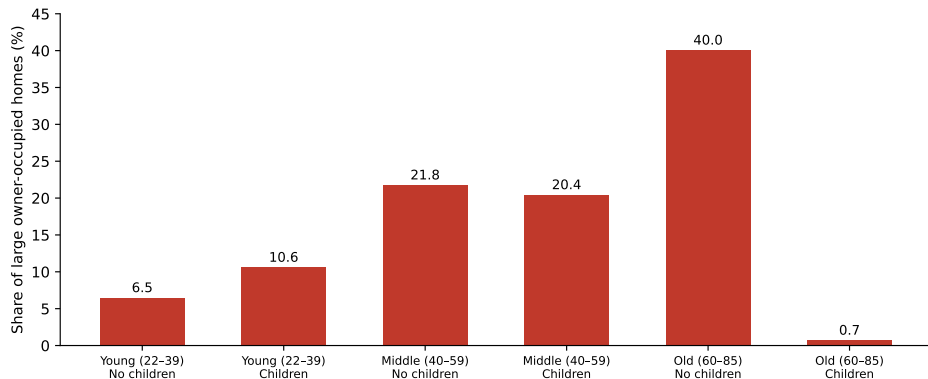
Thank You!

Appendix

The 2023 Equilibrium

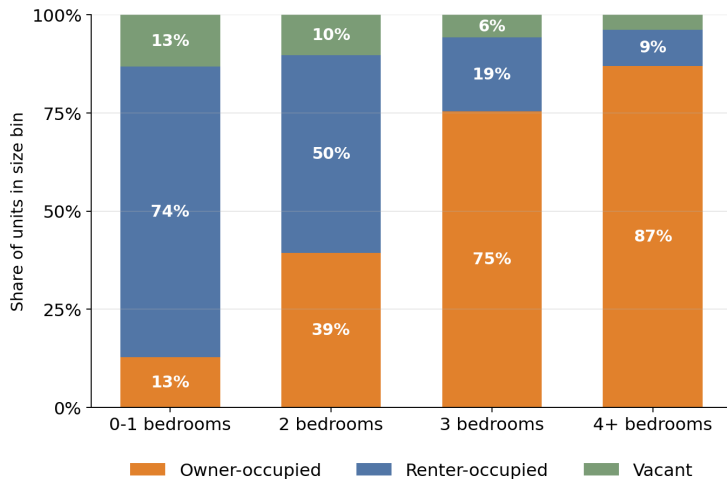


Who Owns the Large Homes



Owner-occupied homes with 6+ rooms; ACS 2023.

Big Units Are Owned

[▶ Back](#)

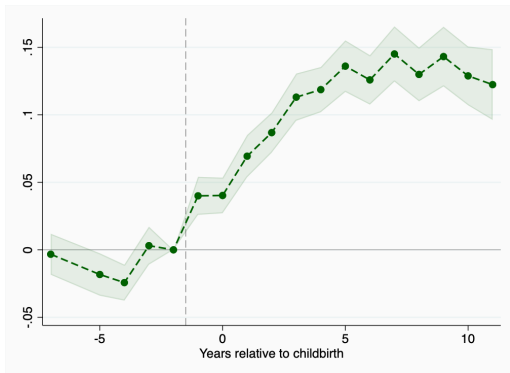
Source: American Housing Survey, 2023.

- **AHS and ACS:** housing sizes, tenure, household composition, and age profiles.
- **PSID:** housing around childbirth, earnings, and wealth by age.
- **CPS and natality records:** completed fertility, childlessness, and birth timing.
- **Census population estimates:** age–sex population and demographic flows.

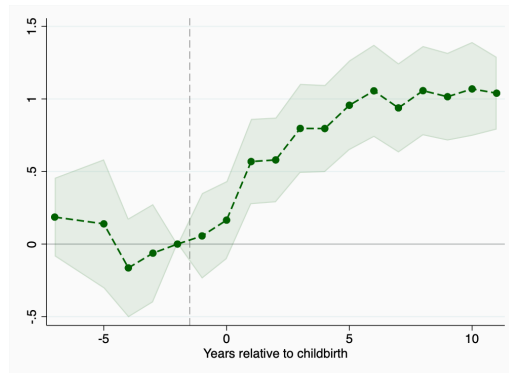
Cross-sectional housing patterns discipline access to space; longitudinal data describe housing adjustment around births.

Ownership and Housing Size Around First Birth

Ownership probability



Number of rooms

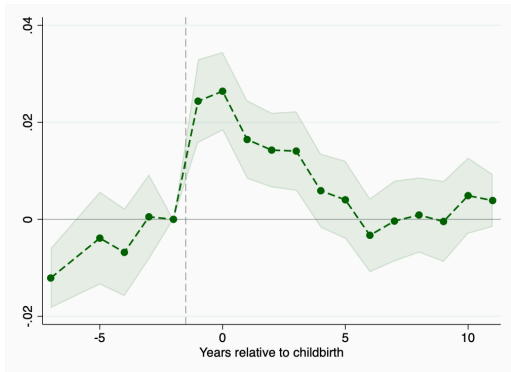


Original May event-study coefficients; measurement revisions under review.

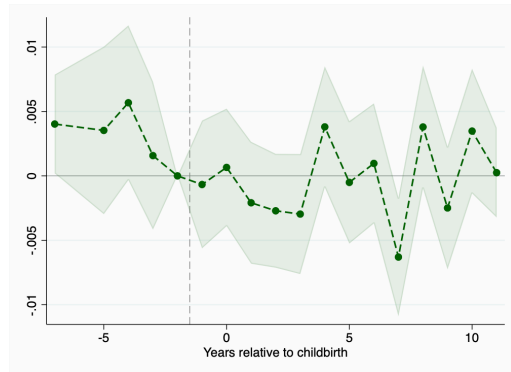
Source: PSID; original May event-study estimates.

Moving Around First Birth

Expansion or better housing



Neighborhood, schools, or proximity



Original May event-study coefficients; measurement revisions under review.

Source: PSID; original May event-study estimates.

Steady-State Solution

Solve for the house price P , pension ϖ , and rebate T , given parameters and a fixed fertility preference.

1. Guess (P, ϖ, T) ; derive rent r from the user cost.
2. Solve households backward over age; compute choice probabilities.
3. Construct the stationary household distribution. Set household mass equal to housing supply divided by mean housing per household.
4. Compute three residuals: housing excess demand, the pension budget, and the rebate budget. The loss is the largest residual in absolute value.
5. Update (P, ϖ, T) and repeat until the loss is below tolerance. Then apply the population law for one period and verify that the distribution reproduces itself.

Transition Solution

Solve for the paths $(P_t, \varpi_t, T_t)_{t \geq t_0}$, given inherited households and a preference believed permanent.

1. Solve the new steady state; it supplies the terminal value function.
2. Guess the three paths; derive rents from the user cost.
3. Solve households backward in calendar time from the terminal value.
4. Advance households and entry queues forward from the inherited state; aggregate housing demand and taxes at each date.
5. Compute the three residuals at every date. The loss is the largest residual across dates.
6. Update the paths jointly and repeat until the loss is below tolerance. Extend the horizon to check that early responses are stable.